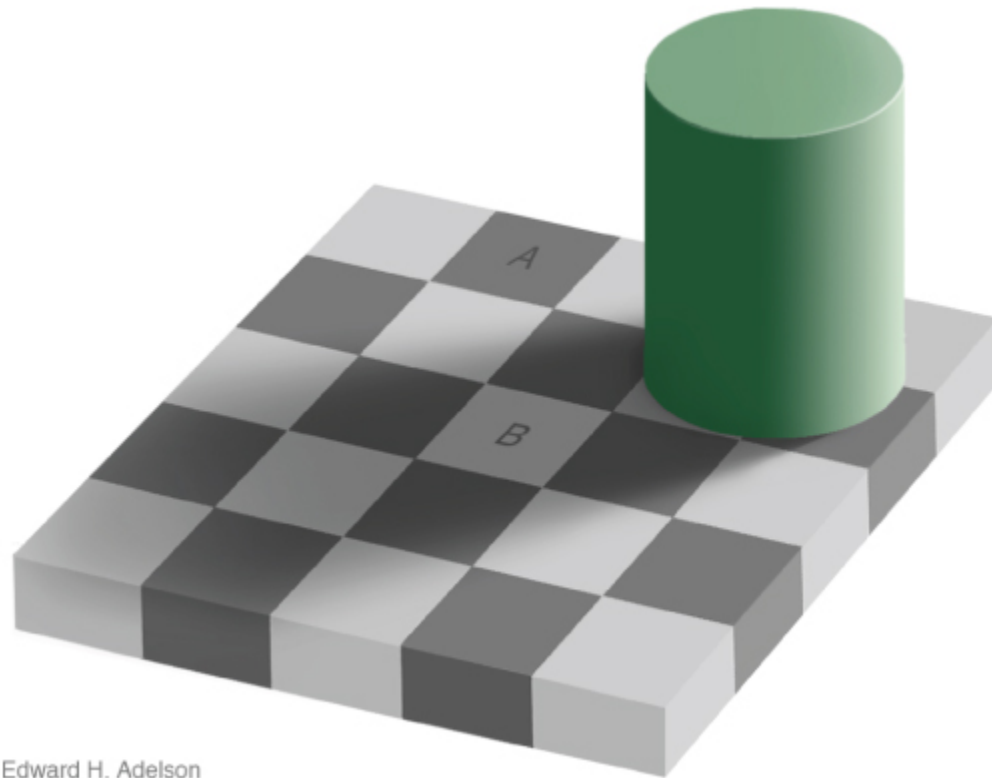




Edward H. Adelson

Figure 1.1 Ed Adelson Checkerboard Illusion

Then ask yourself:

“How much would I bet that A is darker than B?” Would you bet \$5? \$20? \$100?

Or perhaps you would borrow money from a bank, and leverage your bet up 10 times and bet \$1,000,000 on this bet. Why not, right? It is a *guarantee*.

We know how a human approaches this question, but how does a computer think about this question? A computer identifies the red-green-blue (RGB) values for a pixel in box A and the RGB values for a pixel in B. Next, the computer tabulates the results: 120-120-120 for box A; 120-120-120 for box B. Finally, the computer compares the RGB values of the pixel in A and the pixel in B, identifies a match, and concludes that box A and box B are the exact same color. The results are clear to the computer.

So which is it? After taking into consideration the results from the computer algorithm, would you still consider A darker than B? We don't know about you, but we still think A looks darker than B—call us crazy. But then that's what makes us human—we aren't perfect.